

Capstone Supply Chain Management Final Report

FMCG Company

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1. Problem Definition

1.1 Context

A fast-moving consumer goods (FMCG) company that entered the instant noodles market two years ago is facing challenges in balancing demand and supply across its warehouses nationwide. In some regions, demand far exceeds supply, while in others, excess inventory is leading to wastage and higher costs. This mismatch results in significant inventory losses and missed sales opportunities. To remain competitive and profitable in the highly dynamic FMCG sector, the company needs a data-driven approach to optimize its supply chain and align production with regional demand patterns.

1.2 Objective

As a data scientist, you are tasked with building a predictive model using historical sales and supply data to determine the *optimum quantity of product to ship to each warehouse*. Additionally, analyzing demand patterns across different regions will help management design targeted advertisement campaigns. Solving this problem will enable the company to minimize inventory costs, reduce stockouts, and drive higher sales through better demand forecasting and efficient supply chain management.

1.3 Business Opportunity and Supply Chain Optimization

From a **business perspective**, accurate demand planning enables faster response to market changes, improved service levels (higher product availability), lower operational and logistics costs, and better coordination between production, warehouses, and distributors.

Supply Chain Management (SCM) involves the coordination and optimization of procurement, production, warehousing, transportation, and distribution to deliver products efficiently from manufacturer to end consumer. **Supply chain optimization** applies data analytics, predictive modelling, and operational strategies to minimize costs and maximize efficiency across the supply chain.

SCM is important as inefficient supply chains increase operational costs by 15–20% in FMCG companies.

Industry reports show that optimized supply chains can reduce logistics and transportation costs by 10–15%, improve warehouse utilization efficiency by 20–30%, and increase order fulfillment accuracy by 25%.

Companies that use advanced analytics in SCM report 30–40% improvement in overall supply chain performance and faster response time to market demand changes

1.4 Data Description

The data contains the different factors to analyze for the content. The detailed data dictionary is given below.

- Ware_house_ID: Product warehouse ID
- WH_Manager_ID: Employee ID of warehouse manager
- Location_type: Location of the warehouse, like in a city or a village
- WH_capacity_size: Storage capacity size of the warehouse
- zone: Zone of the warehouse

- WH_regional_zone: Regional zone of the warehouse under each zone
- num_refill_req_13m: Number of times refilling has been done in the last 3 months
- transport_issue_11y: Any transport issue, like an accident or goods stolen, reported in the last year
- Competitor_in_mkt: Number of instant noodles competitors in the market
- retail_shop_num: Number of retail shops that sell the product under the warehouse area
- wh_owner_type: Company owns the warehouse, or they have rented the warehouse
- distributor_num: Number of distributors working between the warehouse and retail shops
- flood_impacted: Warehouse is in the flood-impacted area indicator
- flood_proof: Warehouse is flood-proof. Like storage is at some height, not directly on the ground
- electric_supply: The Warehouse has an electric backup, like a generator, so they can run the warehouse in load shedding
- dist_from_hub: Distance between the warehouse to the production hub in Kms
- workers_num: Number of workers working in the warehouse
- wh_est_year: Warehouse established year
- storage_issue_reported_13m: Warehouse reported a storage issue to the corporate office in the last 3 months. Like rats, fungus due to moisture, etc.
- temp_reg_mach: Warehouse has a temperature regulating machine indicator
- approved_wh_govt_certificate: What kind of standard certificate has been issued to the warehouse from the government regulatory body
- wh_breakdown_13m: Number of times the warehouse faced a breakdown in the last 3 months. Like a strike by workers, a flood, or an electrical failure
- govt_check_13m: Number of times government officers have visited the warehouse to check the quality and expiry of stored food in the last 3 months
- product_wg_ton: Product has been shipped in the last 3 months. Weight is in tons

2. Data Overview

Load the required packages, set the working directory, and create the data file.

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not

	Ware_house_ID	WH_Manager_ID	Location_type	WH_capacity_size	zone	WH_regional_zone	num_refill_req_13m	transport_issue_11y	Competitor_in_mkt
0	WH_100000	EID_50000	Urban	Small	West	Zone 6	3	1	2
1	WH_100001	EID_50001	Rural	Large	North	Zone 5	0	0	4
2	WH_100002	EID_50002	Rural	Mid	South	Zone 2	1	0	4
3	WH_100003	EID_50003	Rural	Mid	North	Zone 3	7	4	2
4	WH_100004	EID_50004	Rural	Large	North	Zone 5	3	1	2

5 rows × 24 columns

retail_shop_num	...	electric_supply	dist_from_hub	workers_num	wh_est_year	storage_issue_reported_13m	temp_reg_mach	approved_wh_govt_certificate
4651	...	1	91	29.0	NaN	13	0	A
6217	...	1	210	31.0	NaN	4	0	A
4306	...	0	161	37.0	NaN	17	0	A
6000	...	0	103	21.0	NaN	17	1	A+
4740	...	1	112	25.0	2009.0	18	0	C

Table 1: Top five rows of the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Ware_house_ID                        25000 non-null  object
1   WH_Manager_ID                       25000 non-null  object
2   Location_type                        25000 non-null  object
3   WH_capacity_size                     25000 non-null  object
4   zone                                 25000 non-null  object
5   WH_regional_zone                     25000 non-null  object
6   num_refill_req_l3m                  25000 non-null  int64
7   transport_issue_l1y                 25000 non-null  int64
8   Competitor_in_mkt                   25000 non-null  int64
9   retail_shop_num                     25000 non-null  int64
10  wh_owner_type                       25000 non-null  object
11  distributor_num                      25000 non-null  int64
12  flood_impacted                      25000 non-null  int64
13  flood_proof                         25000 non-null  int64
14  electric_supply                     25000 non-null  int64
15  dist_from_hub                       25000 non-null  int64
16  workers_num                         24010 non-null  float64
17  wh_est_year                         13119 non-null  float64
18  storage_issue_reported_l3m           25000 non-null  int64
19  temp_reg_mach                       25000 non-null  int64
20  approved_wh_govt_certificate         24092 non-null  object
21  wh_breakdown_l3m                    25000 non-null  int64
22  govt_check_l3m                      25000 non-null  int64
23  product_wg_ton                      25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table 2: Basic Information of the Dataset

- There are 25000 rows and 24 columns in the dataset.
- Several identifier and categorical fields ('Ware_house_ID', 'WH_Manager_ID', 'Location_type', 'zone', 'WH_regional_zone', 'wh_owner_type') are correctly stored as object types.
- Count-based columns ('num_refill_req_l3m', 'Competitor_in_mkt', 'retail_shop_num', 'distributor_num', 'wh_breakdown_l3m', 'govt_check_l3m', 'product_wg_ton') are correctly stored as integer types.

2.1 Statistical Summary

Inspecting the Summary Statistics of the Dataset

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Ware_house_ID	25000	25000	WH_100000	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
WH_Manager_ID	25000	25000	EID_50000	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Location_type	25000	2	Rural	22957	NaN	NaN	NaN	NaN	NaN	NaN	NaN
WH_capacity_size	25000	3	Large	10169	NaN	NaN	NaN	NaN	NaN	NaN	NaN
zone	25000	4	North	10278	NaN	NaN	NaN	NaN	NaN	NaN	NaN
WH_regional_zone	25000	6	Zone 6	8339	NaN	NaN	NaN	NaN	NaN	NaN	NaN
num_refill_req_l3m	25000.0	NaN	NaN	NaN	4.08904	2.606612	0.0	2.0	4.0	6.0	8.0
transport_issue_l1y	25000.0	NaN	NaN	NaN	0.77368	1.199449	0.0	0.0	0.0	1.0	5.0
Competitor_in_mkt	25000.0	NaN	NaN	NaN	3.1042	1.141663	0.0	2.0	3.0	4.0	12.0
retail_shop_num	25000.0	NaN	NaN	NaN	4985.71156	1052.825252	1821.0	4313.0	4859.0	5500.0	11008.0
wh_owner_type	25000	2	Company Owned	13578	NaN	NaN	NaN	NaN	NaN	NaN	NaN
distributor_num	25000.0	NaN	NaN	NaN	42.41812	16.064329	15.0	29.0	42.0	56.0	70.0
flood_impacted	25000.0	NaN	NaN	NaN	0.09816	0.297537	0.0	0.0	0.0	0.0	1.0
flood_proof	25000.0	NaN	NaN	NaN	0.05464	0.227281	0.0	0.0	0.0	0.0	1.0
electric_supply	25000.0	NaN	NaN	NaN	0.65688	0.474761	0.0	0.0	1.0	1.0	1.0
dist_from_hub	25000.0	NaN	NaN	NaN	163.53732	62.718609	55.0	109.0	164.0	218.0	271.0
workers_num	24010.0	NaN	NaN	NaN	28.944398	7.872534	10.0	24.0	28.0	33.0	98.0
wh_est_year	13119.0	NaN	NaN	NaN	2009.383185	7.52823	1996.0	2003.0	2009.0	2016.0	2023.0
storage_issue_reported_l3m	25000.0	NaN	NaN	NaN	17.13044	9.161108	0.0	10.0	18.0	24.0	39.0
temp_reg_mach	25000.0	NaN	NaN	NaN	0.30328	0.459684	0.0	0.0	0.0	1.0	1.0
approved_wh_govt_certificate	24092	5	C	5501	NaN	NaN	NaN	NaN	NaN	NaN	NaN
wh_breakdown_l3m	25000.0	NaN	NaN	NaN	3.48204	1.690335	0.0	2.0	3.0	5.0	6.0
govt_check_l3m	25000.0	NaN	NaN	NaN	18.81228	8.632382	1.0	11.0	21.0	26.0	32.0
product_wg_ton	25000.0	NaN	NaN	NaN	22102.63292	11607.755077	2065.0	13059.0	22101.0	30103.0	55151.0

Table 3: Numerical summarization of the dataset

Observation:

- Warehouses require refilling about 4 times in the last 3 months on average, while transport issues are relatively infrequent, with most warehouses reporting zero issues in the past year.
- Risk and infrastructure indicators show that flood-impacted ($\approx 10\%$) and flood-proof ($\approx 5\%$) warehouses are relatively rare. Around 66% of warehouses have electric backup, while only 30% are equipped with temperature-regulating machines.
- In terms of workforce and age, warehouses employ about 29 workers on average, and those with available data were established around 2009. Storage issues are moderately common, with an average of 17 issues reported in the last 3 months, while breakdowns average 3–4 incidents in the same period.
- Shipment volumes are substantial, with an average of $\sim 22,100$ tons of product shipped in the last 3 months, though the wide range indicates significant variation in warehouse throughput.

3. Exploratory Data Analysis

3.1 Univariate

Observation on the Numerical Column

For performing Univariate analysis, we will take a look at the box plots and Histograms to get a better understanding of the distributions

➤ retail_shop_num

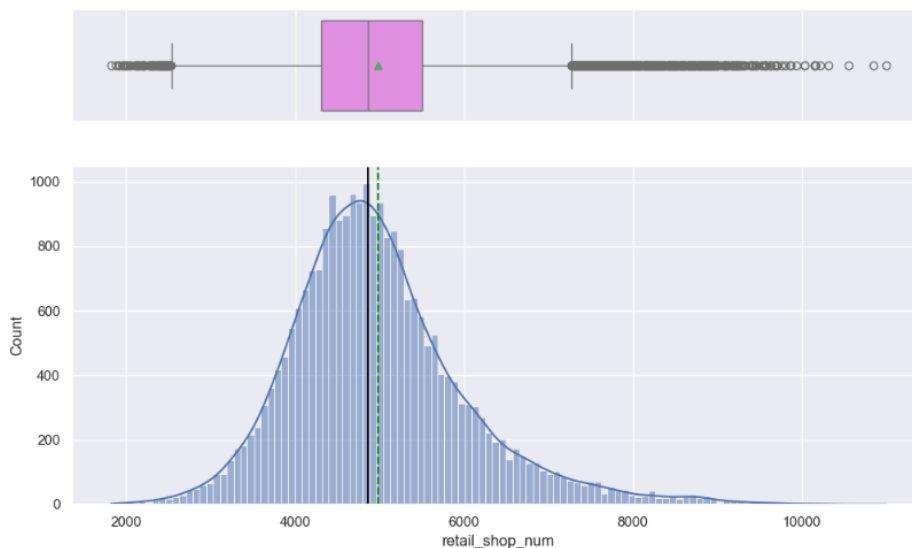


Figure 1: Univariate analysis of retail_shop_num

Observation:

- The plot shows a kind of normal distribution, and there is a larger number of outliers present

➤ product_wg_ton

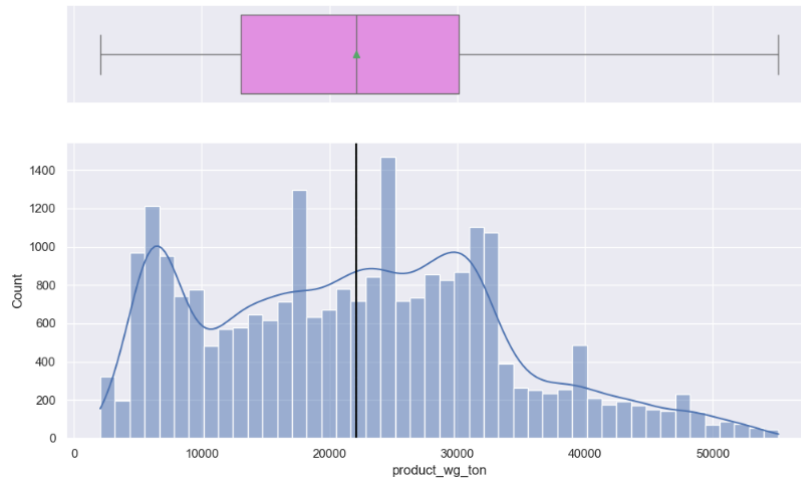


Figure 2: Univariate analysis of product_wg_ton

Observation:

- The **product_wg_ton** distribution is right-skewed with a long tail, indicating most warehouses ship moderate quantities while a few handle very large volumes.

➤ workers_num

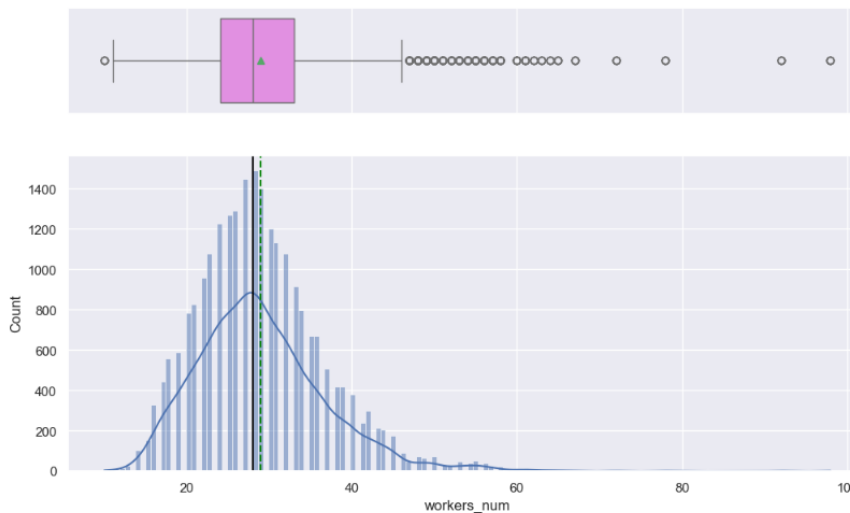


Figure 3: Univariate analysis of workers_num

Observation:

- The number of workers is right-skewed, with most warehouses employing around 20–30 workers and a few warehouses showing high outliers with significantly larger staff sizes.

➤ wh_est_year

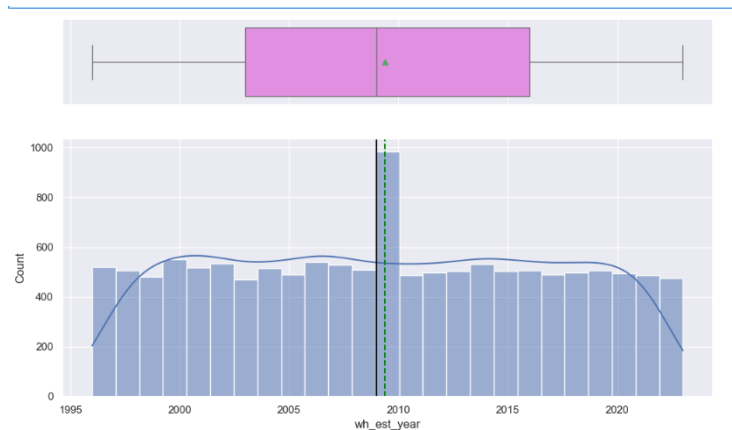


Figure 4: Univariate analysis of wh_est_year

Observation:

- The warehouses are fairly evenly distributed across establishment years, with no strong skewness and a median around the late 2000s.

Overall Observation

The presence of skewness in several independent variables impacts the objective of establishing reliable relationships with the target variable **product_wg_ton** by distorting mean-based interpretation and weakening linearity assumptions. Right-skewed variables, such as **retail_shop_num** and **workers_num**, are influenced by a small number of high-volume warehouses, which can distort mean-based relationships with **product_wg_ton**.

Observation on the Categorical Column

➤ Location_type

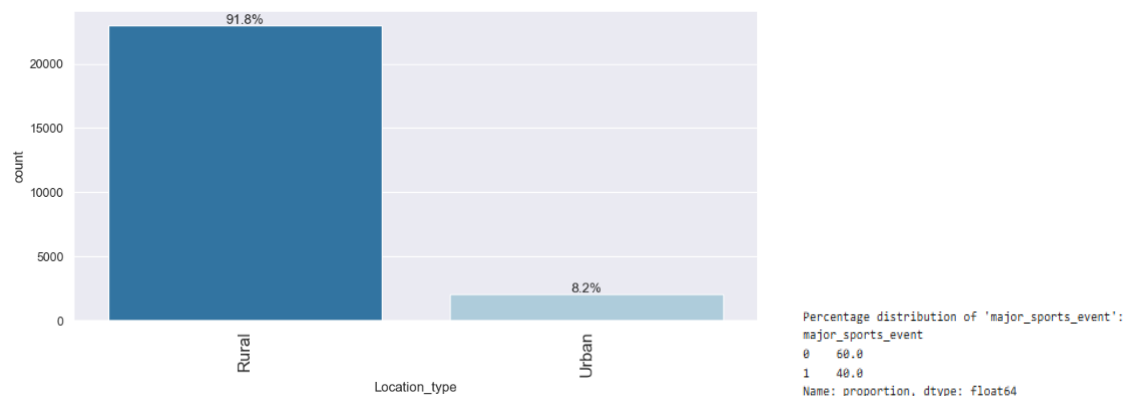


Figure 5: Univariate analysis of Location_type

Observation:

- It was observed that 91.8% of the dataset has rural locations.

➤ WH_capacity_size

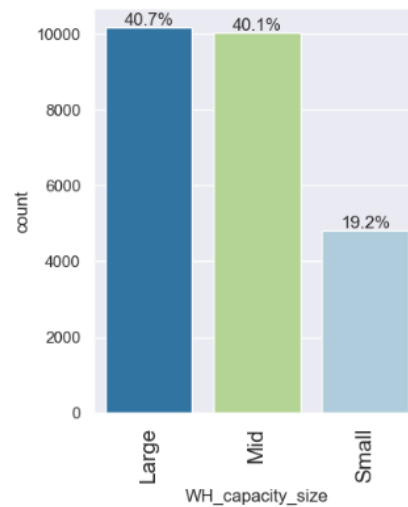


Figure 6: Univariate analysis of WH_capacity_size

Observation:

- It was observed that Large and Mid have approx. 40% capacity

➤ num_refill_req_l3m

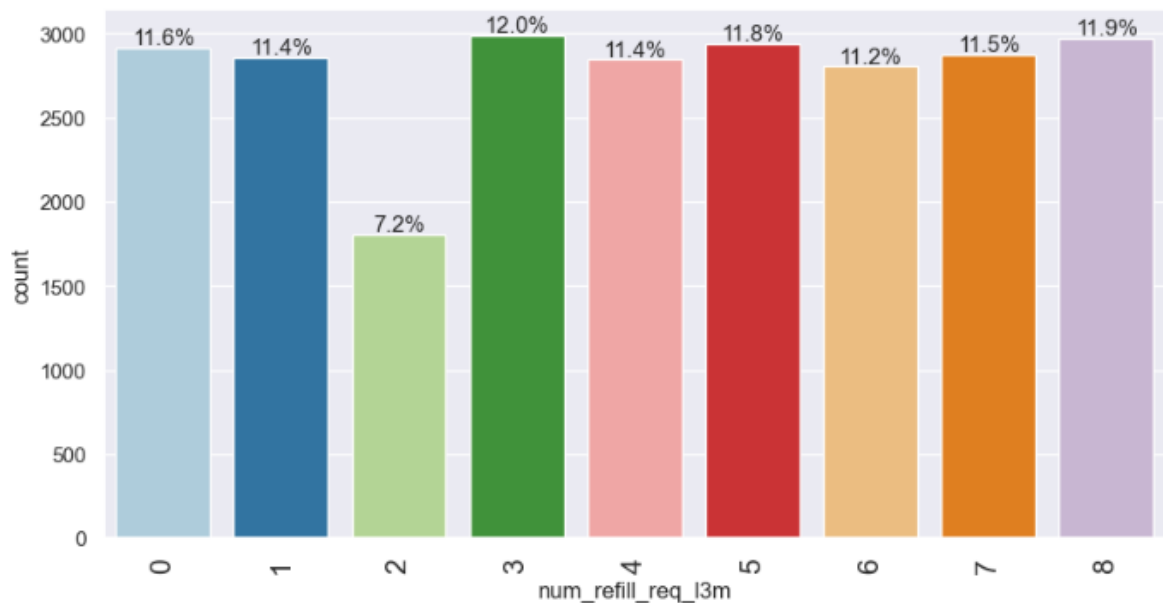


Figure 7: Univariate analysis of num_refill_req_l3m

Observation:

- The distribution shows that most of the refills are from 3, which is approx. 12%

➤ Competitor_in_mkt

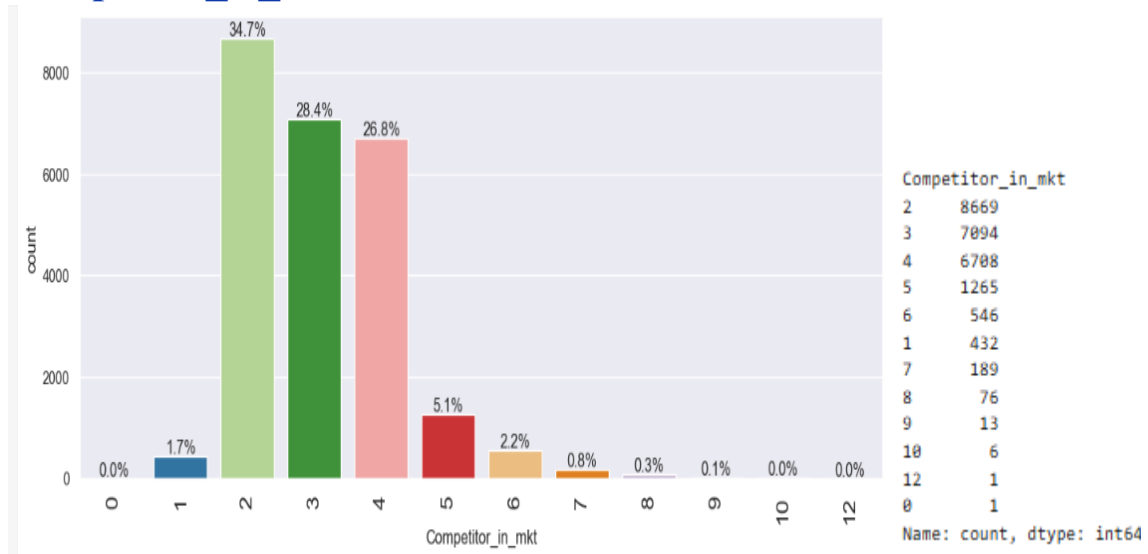


Figure 8: Univariate analysis of Competitor_in_mkt

Observation:

- It was observed that 2 has lots of competitors, as 0 only has 1

➤ approved_wh_govt_certificate

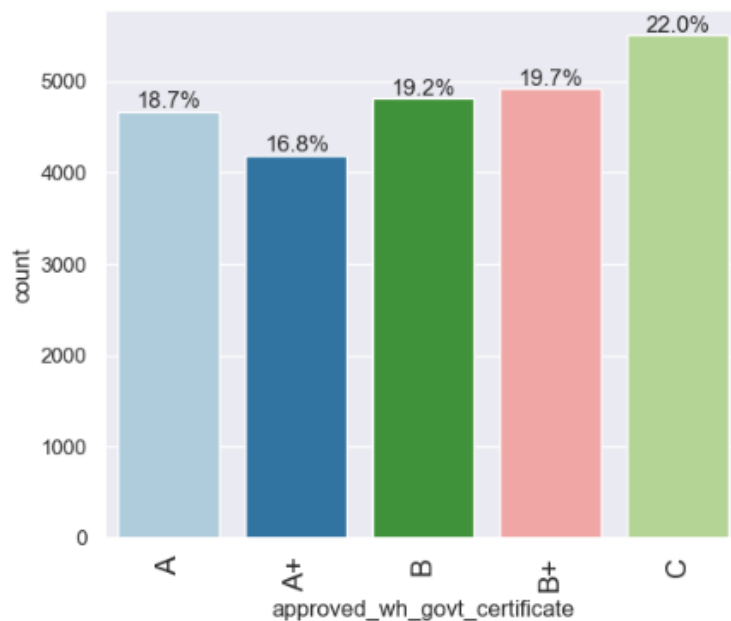


Figure 9: Univariate analysis of approved_wh_govt_certificate

Observation:

- It was observed that C certified has a larger amount compared to A+.

➤ wh_breakdown_l3m

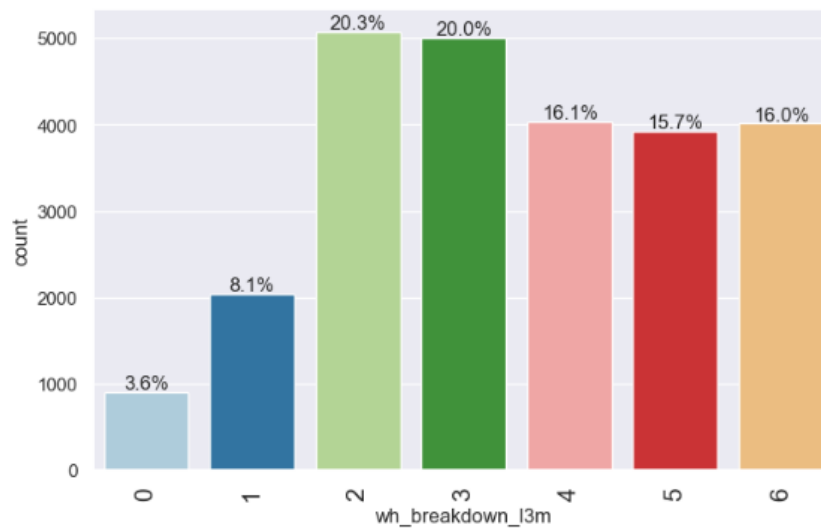


Figure 10: Univariate analysis of wh_breakdown_l3m

Observation:

- It was observed that 2 and 3 have approx. 20% distribution.

Mean-medium base analysis:

Group-wise statistics for Location_type				Group-wise statistics for electric_supply			
	count	mean	median		count	mean	median
Location_type				electric_supply			
1	2043	25004.033774	25074.0	1	16422	22084.513884	22105.0
0	22957	21844.430108	22072.0	0	8578	22137.320588	22092.0
Group-wise statistics for WH_capacity_size				Group-wise statistics for flood_impacted			
	count	mean	median		count	mean	median
WH_capacity_size				flood_impacted			
2	10020	22202.298104	22128.5	0	22546	22111.436707	22103.0
3	10169	22100.487855	22093.0	1	2454	22021.748574	22089.5
1	4811	21899.591561	22066.0				
Group-wise statistics for wh_owner_type				Group-wise statistics for flood_proof			
	count	mean	median		count	mean	median
wh_owner_type				flood_proof			
0	11422	22176.125810	22118.0	1	1366	22081.318448	22104.0
1	13578	22040.809692	22086.5	0	23634	22103.864856	22101.0

Table 4: Mean-medium analysis of categorical variables

Group-wise mean and median analysis indicate that not all categorical variables meaningfully separate the target variable *product_wg_ton*. **Location type** exhibits clear and consistent separation, identifying urban warehouses as higher-demand hubs. In contrast, variables such as **warehouse ownership**, **electric**

supply, and flood-proofing show negligible differences, suggesting limited direct impact on shipment volume. Risk-related attributes, such as *flood impact*, introduce mild variability but do not act as primary demand drivers. These insights helped prioritize variables based on practical and predictive relevance rather than inclusion by default.

3.2 Multivariate

Correlation of Numerical Variables

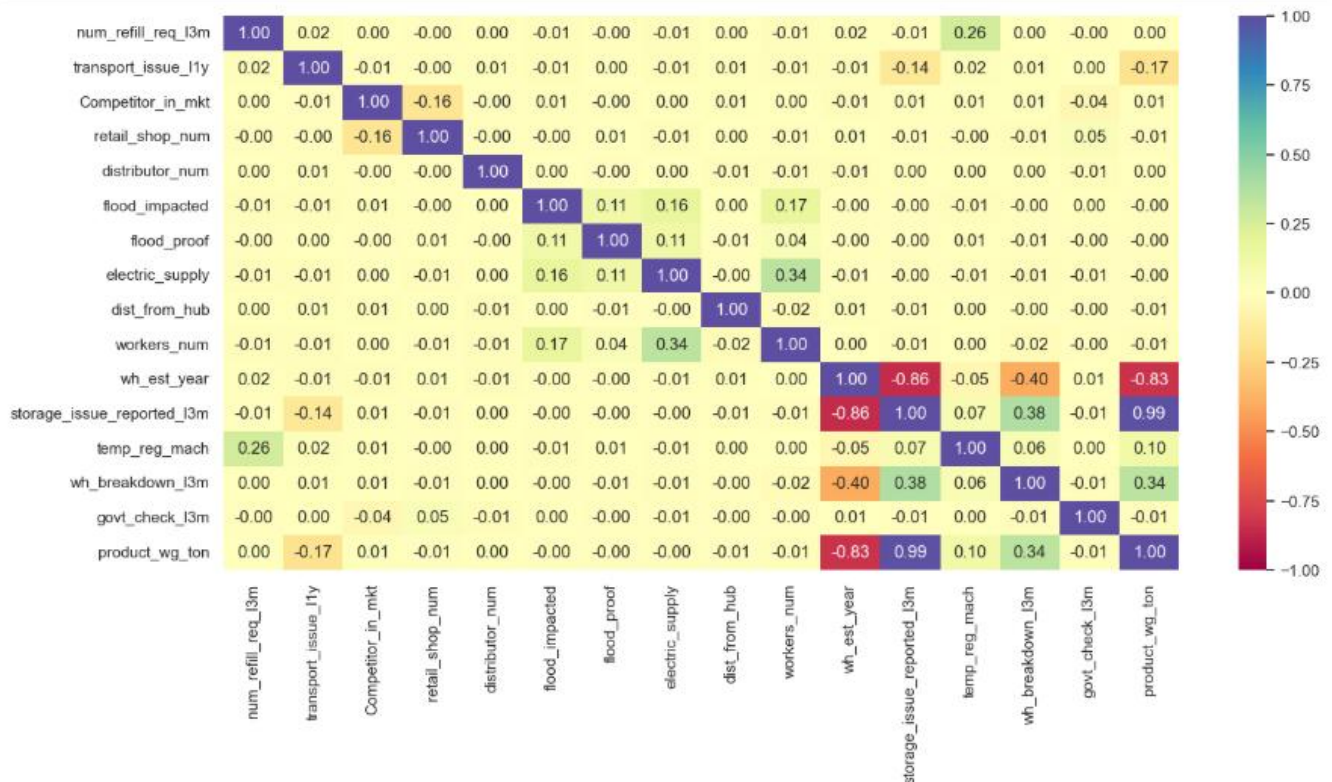


Figure 11: Correlation of numerical variables

Observation:

- Product shipment volume is tightly linked with storage issues and warehouse breakdowns, indicating higher operational stress at high-throughput warehouses.
- Older warehouses tend to handle larger shipment volumes and report more storage-related issues compared to newer facilities.
- Infrastructure readiness (electric supply and workforce size) shows moderate alignment, suggesting better-equipped warehouses employ more staff.
- Refill frequency has a mild association with temperature-regulating machines, reflecting higher activity levels in controlled-storage facilities.
- Market factors (competitors, retail shops, distributors) and external checks (government inspections, flood indicators) show minimal direct correlation with operational performance.

3.3 Bivariate

Relationship between Numerical & Categorical Attributes with the target variable

➤ retail_shop_num vs product_wg_ton

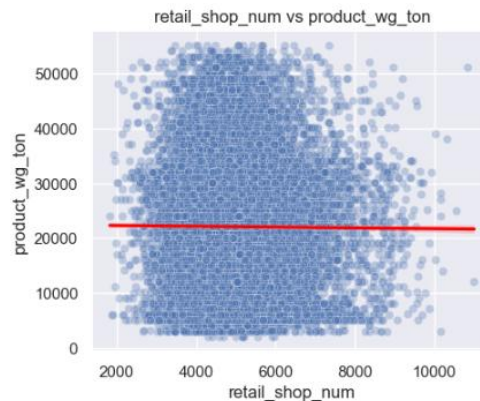


Figure 12: retail_shop_num vs product_wg_ton

- The scatter plot shows a dense cloud with a nearly flat regression line. This suggests that although retail coverage is necessary to meet demand, an increase in the number of retail shops does not proportionally increase shipment volume. Hence, retail_shop_num has a **non-linear and indirect impact** on product_wg_ton.

➤ storage_issue_reported_l3m vs product_wg_ton

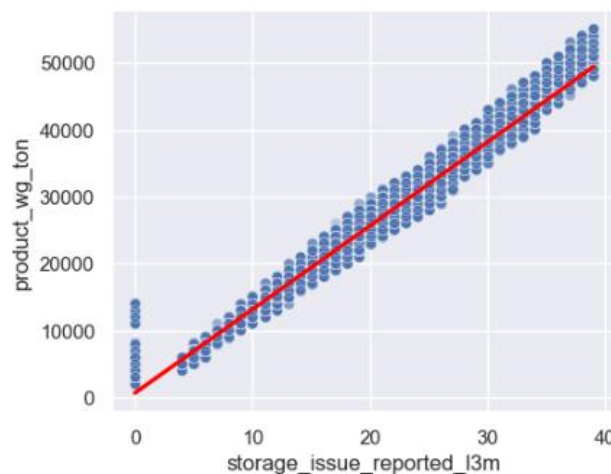


Figure 13: storage_issue_reported_l3m vs product_wg_ton

- A strong, clear positive linear relationship is visible, with shipment volume increasing steadily as storage issues increase. This strong association indicates that higher shipment volumes place operational stress on warehouses, leading to more storage issues. This variable shows a **very strong relationship** with product_wg_ton.

➤ **wh_breakdown_l3m vs product_wg_ton**

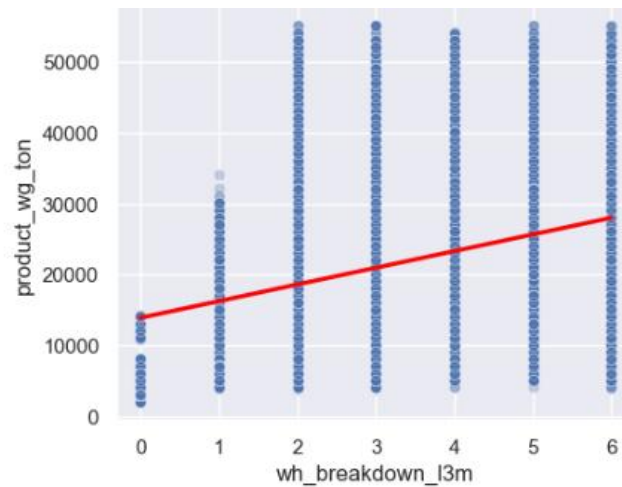


Figure 14: wh_breakdown_l3m vs product_wg_ton

- Shipment volumes increase as the number of breakdowns rises, with a clear positive regression trend. wh_breakdown_l3m reflects **operational stress associated with high shipment activity**, showing a **moderate to strong positive relationship** with product_wg_ton

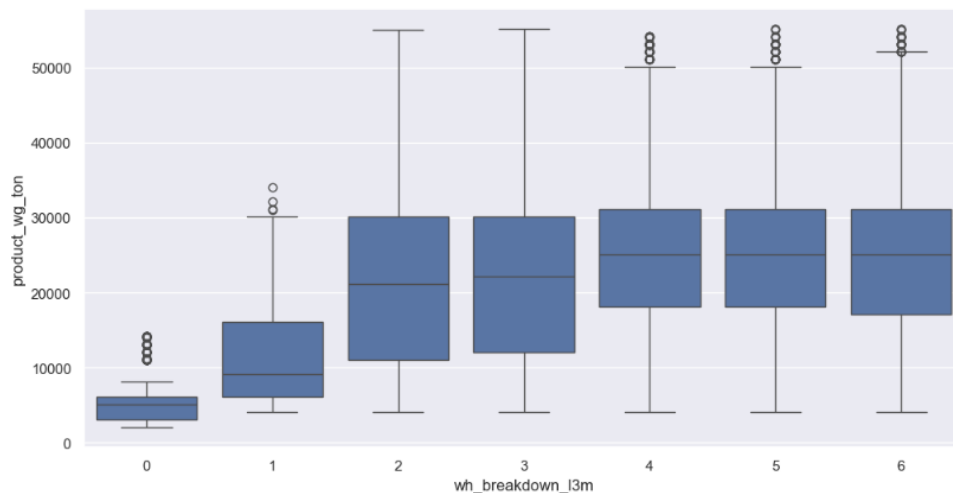


Figure 15: wh_breakdown_l3m vs product_wg_ton

- Product shipment volume generally increases with the number of warehouse breakdowns, indicating that higher-activity warehouses tend to experience more breakdown incidents.

➤ **storage_issue_reported_l3m vs product_wg_ton**

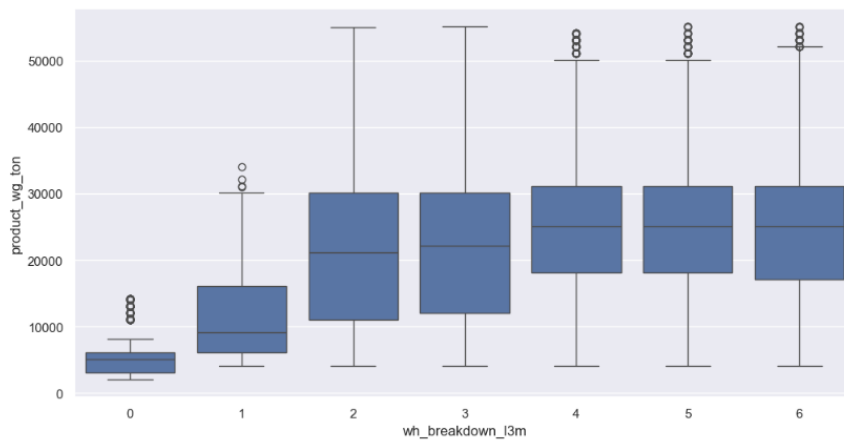


Figure 16: storage_issue_reported_l3m vs product_wg_ton

- Product shipment volume increases steadily with the number of storage issues reported, indicating a strong positive relationship between warehouse activity level and reported storage problems.

➤ product_wg_ton vs wh_est_year

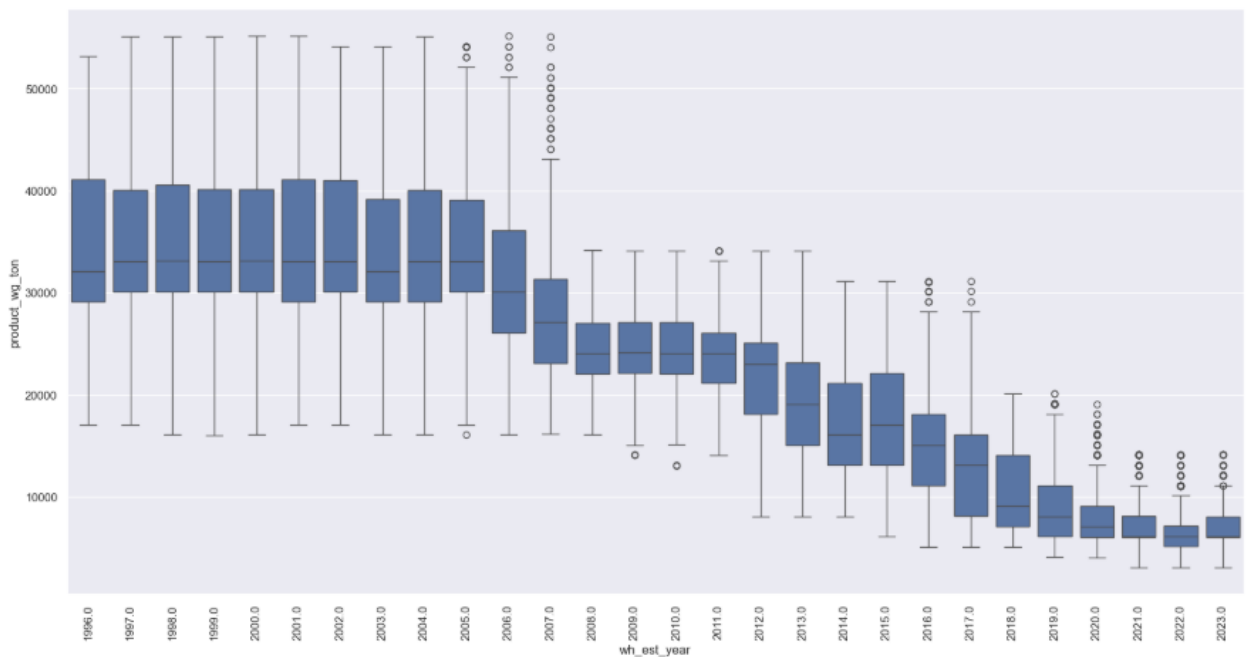


Figure 17: product_wg_ton vs wh_est_year

- Product shipment volume tends to be higher for older warehouses and gradually decreases for more recently established warehouses, indicating a negative relationship between establishment year and shipped quantity. This indicates a **strong negative relationship** between establishment year and shipment volume, making wh_est_year an **important explanatory variable** for product_wg_ton

Overall EDA Observations

- Operational stress indicators, particularly storage issues and warehouse breakdowns, show the strongest positive relationship with shipment volume, highlighting activity-driven operational pressure.
- Warehouse maturity matters, as older warehouses consistently handle higher shipment volumes compared to recently established facilities.
- Location type differentiates demand, with **urban** warehouses acting as higher-demand hubs, while other categorical factors show limited direct impact.
- Retail coverage and market variables exhibit weak or non-linear relationships with shipment volume, indicating indirect influence rather than proportional demand growth.
- Overall, dynamic operational factors dominate shipment behaviour, while static structural and geographic attributes act as secondary or contextual variables.

4. Data Preprocessing

4.1 Duplicates & Missing values

There are 0 duplicates values

Ware_house_ID	WH_Manager_ID	Location_type	WH_capacity_size	zone	WH_regional_zone	num_refill_req_13m	transport_issue_11y	Competitor_in_mkt	retail_shop_nu
---------------	---------------	---------------	------------------	------	------------------	--------------------	---------------------	-------------------	----------------

0 rows × 24 columns

Table 5: Duplicates Values

Ware_house_ID	0
WH_Manager_ID	0
Location_type	0
WH_capacity_size	0
zone	0
WH_regional_zone	0
num_refill_req_13m	0
transport_issue_11y	0
Competitor_in_mkt	0
retail_shop_num	0
wh_owner_type	0
distributor_num	0
flood_impacted	0
flood_proof	0
electric_supply	0
dist_from_hub	0
workers_num	990
wh_est_year	11881
storage_issue_reported_13m	0
temp_reg_mach	0
approved_wh_govt_certificate	908
wh_breakdown_13m	0
govt_check_13m	0
product_wg_ton	0
dtype: int64	

Table 6: Missing Values

there are missing values in `workers\_num`, `wh\_est\_year`, and `approved\_wh\_govt\_certificate`
Hence we are treating null values by imputing median and encode the value

1. workers_num:

Imputes the median value for null values

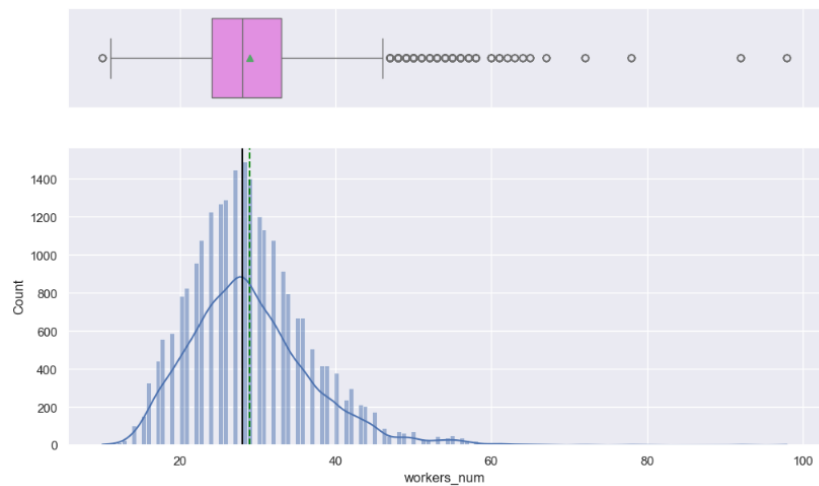


Figure 18: Before median imputing `workers_num`

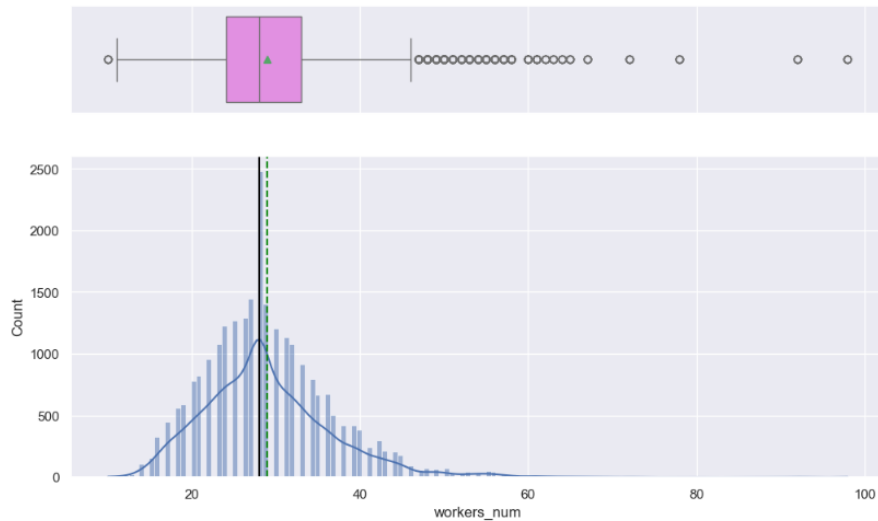


Figure 19: After median imputing `workers_num`

2. `wh_est_year`

Here, converting the `'wh_est_year'` to `'warehouse age'` as it will be more suitable for modelling and analyses, and removing the year column from the dataset

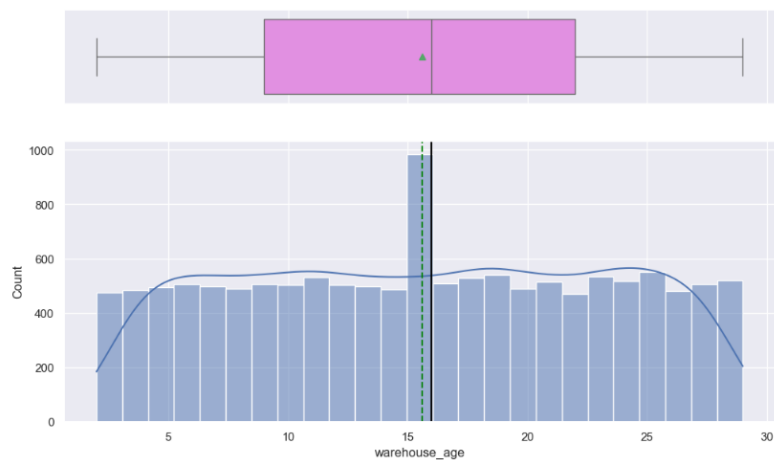


Figure 20: convert from `wh_est_year` to `warehouse age`

The warehouse establishment year was transformed into warehouse age to better capture its operational relevance, as age directly reflects infrastructure maturity and degradation effects.

Now Perform correlation / regression significance test with target.

corr, p_value = (0.829, 0.000e+00)

A warehouse age showed a strong and significant association with shipment volume ($p < 0.001$) and was correlated with multiple capacity and stability indicators. KNN imputation was used to preserve local similarity and structural relationships.

KNN imputation was applied to warehouse_age because the variable is numerically continuous and exhibits meaningful relationships with other operational attributes. Unlike mean or median imputation, KNN preserves local similarity by imputing missing values based on comparable warehouses, thereby maintaining structural relationships and improving model reliability.

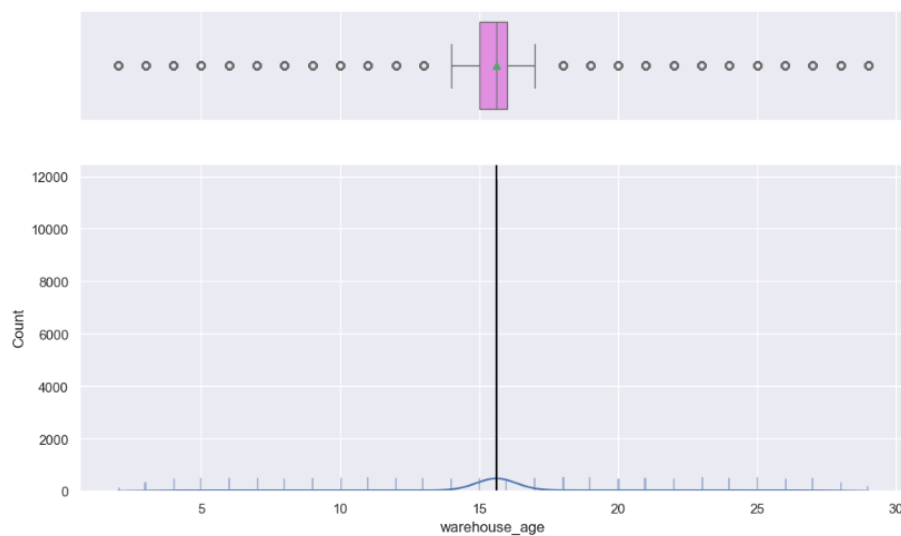


Figure 21: After KNN imputing warehouse_age

3. approved_wh_govt_certificate

We will use a value mode impute method to fill missing values for the approved_wh_govt_certificate columns, as it is a categorical attribute.

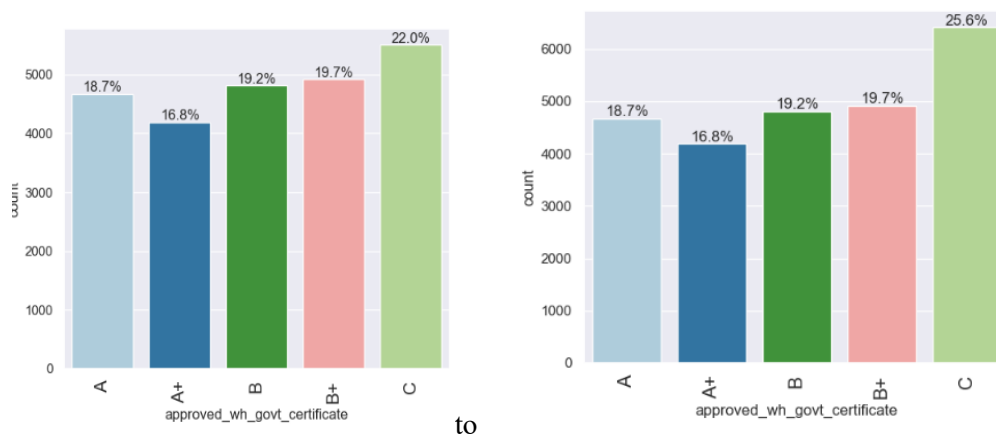


Figure 22: Mode imputing approved_wh_govt_certificate

```
Ware_house_ID      0
WH_Manager_ID      0
Location_type       0
WH_capacity_size    0
zone               0
WH_regional_zone    0
num_refill_req_13m  0
transport_issue_11y 0
Competitor_in_mkt  0
retail_shop_num     0
wh_owner_type       0
distributor_num     0
flood_impacted      0
flood_proof         0
electric_supply      0
dist_from_hub       0
workers_num         0
storage_issue_reported_13m 0
temp_reg_mach       0
approved_wh_govt_certificate 0
wh_breakdown_13m    0
govt_check_13m      0
product_wg_ton      0
warehouse_age       0
dtype: int64
```

- All missing values are imputed

Table 7: after treating missing values

4.2 Outliers

Let's find out the outliers in the numerical attributes

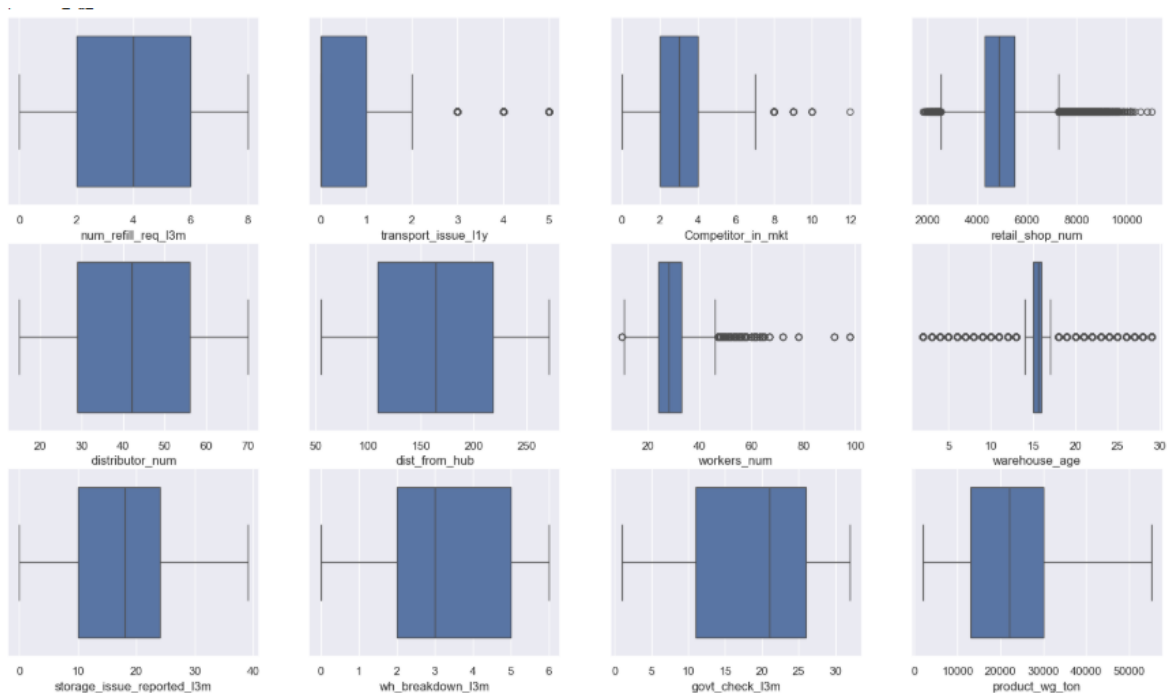


Figure 23: Outliers of numerical attributes

There are various outliers in the *workers_num*, *retail_shop_num*, but these outliers can be considered valid. They won't be removed from the data set, as there are some cases where the retail shop has high number and worker number can be more. Hence, these outliers are considered informative and not treated, and kept in the data for further insights.

4.3 Feature Engineering

approved_wh_govt_certificate, *Location_type*, *wh_owner_type* columns can be encoded to numbers hence we are encode it to number value

```

from: approved_wh_govt_certificate
C      5501
B+     4917
B      4812
A      4671
A+     4191
Not Available  908
Name: count, dtype: int64
-----
Location_type
Rural    22957
Urban    2043
Name: count, dtype: int64
-----
wh_owner_type
Company Owned  13578
Rented        11422
Name: count, dtype: int64

to: approved_wh_govt_certificate
5      5501
3      4917
4      4812
2      4671
1      4191
0       908
Name: count, dtype: int64
-----
Location_type
0      22957
1       2043
Name: count, dtype: int64
-----
wh_owner_type
1      13578
0      11422
Name: count, dtype: int64

```

Table 8: Encoding scaling

Hence, the datatype of this attribute has been changed.

Significant test for categorical variables with the target variable

	Variable	p_value	Significant (p < 0.05)
0	Location_type	3.639843e-32	Yes
1	WH_capacity_size	3.310533e-01	No
2	zone	3.453489e-01	No
3	WH_regional_zone	3.339541e-01	No
4	wh_owner_type	3.585434e-01	No
5	electric_supply	7.327428e-01	No
6	flood_impacted	7.162490e-01	No
7	flood_proof	9.443548e-01	No
8	temp_reg_mach	2.126171e-58	Yes
9	approved_wh_govt_certificate	1.550049e-306	Yes

Table 9: Significant test with target variable

ANOVA results indicate that not all categorical variables significantly differentiate shipment volume. *Location type, temperature regulation availability, and government certification status* exhibit statistically significant differences in **product_wg_ton**, highlighting their importance as demand and capacity differentiators. In contrast, variables related to *ownership, power backup, flood risk, and regional classification* do not show independent statistical significance, suggesting they act as operational or contextual factors rather than direct demand drivers. Accordingly, significant variables were emphasized during feature selection, while non-significant variables were retained selectively to support model stability.

4.4 Data Preparation for Modelling

Creating a dummy dataset for modelling

- Target variable is “product_wg_ton”
- We are scaling the dataset and removing the not useful columns like Ware_house_ID, WH_Manager_ID and applied LabelEncoder to Location_type, WH_capacity_size, approved_wh_govt_certificate and deleted it from both train and test data.
- applied StandardScaler to the numerical variable to scale the values and after scaling data is Train, and Test dataset (row,column)--((17500, 30), (7500, 30))
- Split the data in a 70:30 ratio for train to test data.
- Created x_train dataset, and the Number of rows in the train data is 17500.

- Created `x_test` dataset, and the Number of rows in the test data is 7500

5. Model Building

5.1 Model Building - Baseline

We will perform the LinearRegression Algorithm by using `grid_cv`

For that we need to check the model performance.

- We will check the model performance on the actual weight and not the log values.
- We will be using metric functions defined in sklearn for RMSE and MAE.
- We will be using Mean Absolute Percentage Error (MAPE) is chosen as the primary evaluation metric because it directly aligns with the business objective of optimizing shipment quantities across warehouses.
 - The mean absolute percentage error (MAPE) measures the accuracy of predictions as a percentage, and can be calculated as the average absolute percent error for each predicted value minus actual values divided by actual values.
 - It expresses prediction error in percentage terms, making it easy to interpret for business stakeholders.
 - It allows fair comparison across warehouses of different sizes, unlike absolute metrics.
 - In supply chain planning, relative error is more actionable than absolute error when deciding buffer stock and shipment volumes.

```
Evaluation Result

LinearRegression

mean square error = 2664833.679358074
mean absolute error = 1236.7044914752396
r2 score = 97.97114614814

MAPE = 10.396462207899257
```

After running the model, these are the result:

We are using **MAPE** as the metric of choice

Hence, the Linear Regression model demonstrates strong predictive performance, with an R^2 score of 97.97%, indicating that it explains almost all the variability in `product_wg_ton` across warehouses. The MAE of approximately 1,237 tons suggests that the average prediction error is operationally manageable for supply planning, while the MSE reflects a few larger errors in high-volume warehouses.

Most importantly, the model achieves a MAPE of 10.4%, meaning that shipment quantity predictions are, on average, within $\pm 10\%$ of actual values. This level of percentage error is considered acceptable and reliable in FMCG demand forecasting, making the model well-suited for optimizing warehouse-level shipments, reducing inventory mismatches, and supporting data-driven supply chain decisions.

Final Outcome

The current Linear Regression model shows strong performance with a low MAPE, but it can be improved by reducing multicollinearity using regularized models such as Ridge or Lasso and by handling outliers more effectively. Additional feature engineering, including interaction terms and lag-

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based demand variables, can help capture operational and demand patterns more accurately.

Further gains may come from using non-linear models like Random Forest or Gradient Boosting, which better capture leaving demand relationships. Incorporating seasonality, promotions, and regional factors, along with robust validation and periodic retraining, will improve accuracy and ensure long-term model reliability. So, we are moving forward to other algorithms.

5.2 Model Building - Advanced Models

To improve the model-building process, we will build the Ensemble models listed below:

- Decision Tree
- Bagging
- Random Forest
- Adaboost
- Gradient Boosting
- XGBoost

After these models are built, here are the training and testing performance of the models

Training performance comparison:

	Decision Tree	Bagging Regressor	Random Forest Estimator	Adaboost Regressor	Gradient Boost Estimator	XGB
RMSE	0.0	404.828820	340.808738	1746.039737	882.602453	590.103328
MAE	0.0	269.893966	248.813205	1401.447034	665.842079	449.485199
R-squared	1.0	0.998786	0.999140	0.977418	0.994230	0.997421
Adj. R-squared	1.0	0.998784	0.999138	0.977380	0.994220	0.997416
MAPE	0.0	1.642384	1.536432	10.261142	4.128908	2.736879

Table 10: Training Performance Comparison

Testing performance comparison:

	Decision Tree	Bagging Regressor	Random Forest Estimator	Adaboost Regressor	Gradient Boost Estimator	XGB
RMSE	1248.038586	937.487548	899.307058	1751.022021	889.960281	894.813842
MAE	821.466800	689.098200	664.788599	1408.847773	671.651857	657.914429
R-squared	0.988385	0.993446	0.993969	0.977135	0.994094	0.994029
Adj. R-squared	0.988338	0.993420	0.993945	0.977044	0.994070	0.994005
MAPE	5.117805	4.279265	4.163342	10.376517	4.221128	4.232030

Table 11: Testing Performance Comparison

Observation:

- The **Decision Tree** model achieves perfect performance on the training data ($R^2 = 1.0$, MAPE = 0.0), but its test MAPE increases to approximately 5.12%, indicating strong overfitting due to unrestricted tree growth.
- The **Bagging Regressor** shows excellent training performance (MAPE $\approx 1.64\%$) and improved generalization on the test set (MAPE $\approx 4.28\%$), demonstrating effective variance reduction compared to a single decision tree.
- The **Random Forest** model achieves near-perfect training performance (MAPE $\approx 1.54\%$) and

strong test performance (MAPE $\approx$ 4.16%), indicating a good balance between model complexity and generalization.

- The **AdaBoost Regressor** exhibits comparatively weaker performance, with higher training and testing MAPE values ($\approx$ 10.26% and $\approx$ 10.38%), suggesting limited suitability for capturing the underlying demand patterns in this dataset.
- The **Gradient Boosting** model delivers consistent performance, with low training error (MAPE $\approx$ 4.13%) and stable test performance (MAPE $\approx$ 4.22%), reflecting good generalization.
- The **XGBoost** model performs strongly, achieving low training MAPE ($\approx$ 2.74%) and competitive test MAPE ($\approx$ 4.23%), highlighting its effectiveness in modeling complex non-linear relationships.

Among all the models, **Random Forest**, **Gradient Boosting**, and **XGBoost** demonstrate the most reliable and stable performance on unseen data, making them strong candidates for further tuning and deployment in warehouse-level shipment forecasting.

5.3 Model Performance Improvement using Hyperparameter Tuning

Tuned all the models by different parameters and performed the modelling,

Model Comparison at tuned models

Training performance comparison:

	Decision Tree Tuned	Bagging Regressor Tuned	Random Forest Tuned	Adaboost Tuned	Gradient Boost Tuned	XGB Tuned	Stacking Regressor
RMSE	938.643393	598.214659	755.139040	1688.999215	920.903537	641.296636	837.312516
MAE	694.088168	457.593109	553.681339	1338.489905	700.617415	488.052338	611.901550
R-squared	0.993474	0.997349	0.995776	0.978870	0.993718	0.996954	0.994807
Adj. R-squared	0.993463	0.997345	0.995769	0.978833	0.993708	0.996949	0.994798
MAPE	4.084567	3.028368	3.358721	9.539908	4.436919	2.980200	3.841823

Table 12: Training Performance comparison of tuned model

Testing performance comparison:

	Decision Tree Tuned	Bagging Regressor Tuned	Random Forest Tuned	Adaboost Tuned	Gradient Boost Tuned	XGB Tuned	Stacking Regressor
RMSE	963.721445	1248.604794	883.568664	1683.336899	933.969486	886.595913	907.434984
MAE	715.201126	975.921240	655.058486	1336.918732	712.997376	650.724426	669.522156
R-squared	0.993074	0.988374	0.994178	0.978869	0.993495	0.994138	0.993859
Adj. R-squared	0.993046	0.988327	0.994155	0.978784	0.993469	0.994115	0.993835
MAPE	4.289911	6.592610	4.053796	9.572115	4.582432	4.166632	4.268217

Table 13: Testing Performance comparison of tuned model

Observation:

- The **Tuned Decision Tree** model shows controlled training performance (R^2 - 0.99, MAPE - 4.08%) and stable test performance (MAPE - 4.29%), indicating that hyperparameter constraints successfully reduced overfitting and improved generalization.

- The **Tuned Bagging Regressor** achieves strong training accuracy (MAPE - 3.03%), but its test MAPE increases to approximately 6.59%, suggesting a loss of generalization due to over-regularization.
- The **Tuned Random Forest** model maintains a good balance between training and testing performance, with test MAPE around 4.05% and closely aligned R^2 and Adjusted R^2 values, indicating stable and reliable generalization.
- The **Tuned AdaBoost Regressor** continues to underperform compared to other tuned models, with high training and testing MAPE values (9.54% and 9.57%), confirming limited suitability for this regression problem.
- The **Tuned Gradient Boosting** model delivers consistent results, with test MAPE around 4.58% and minimal divergence between training and testing metrics, reflecting controlled learning and reasonable generalization.
- The **Tuned XGBoost** model demonstrates strong and stable performance, achieving a test MAPE of approximately 4.17% and high R^2 values, making it one of the top-performing tuned models.
- The **Stacking Regressor** performs competitively with a test MAPE of approximately 4.27%, but does not provide a significant improvement over the best individually tuned ensemble models, indicating limited added value from stacking.

Among the tuned models, **Random Forest Tuned** and **XGBoost Tuned** exhibit the most reliable and balanced performance on unseen data, achieving low error rates and strong generalization, and are therefore well-suited for deployment in warehouse-level shipment forecasting.

6. Model Performance Comparison and Final Model Selection

In line with the requirement, multiple advanced regression models were developed, including Decision Tree, Bagging Regressor, Random Forest, AdaBoost, Gradient Boosting, XGBoost, and a Stacking Regressor. Both base and tuned versions of these models were evaluated to assess performance improvement and control overfitting.

Model performance was compared using multiple evaluation metrics—RMSE, MAE, R^2 , Adjusted R^2 , and Mean Absolute Percentage Error (MAPE). MAPE was selected as the primary metric due to its direct alignment with the business objective of minimizing relative shipment forecasting error across warehouses of varying scale.

Final Model Selection

- Based on comparative evaluation across multiple metrics, the **tuned Random Forest model** is selected as the final model.
- The tuned Random Forest achieves:
 - **Lowest test MAPE (approx. 4.05%)**, aligning well with the business objective of minimizing relative shipment prediction error
 - **Low RMSE and MAE**, indicating controlled absolute prediction errors
 - **High and closely aligned R^2 and Adjusted R^2 (~0.994)**, confirming strong explanatory power without overfitting
- These results indicate that the tuned Random Forest model offers the best balance between predictive accuracy, robustness, and interpretability for warehouse-level shipment forecasting.

Although XGBoost and Gradient Boosting also delivered competitive results, Random Forest was selected as the final model due to its marginally better error metrics, stability across train and test datasets, and comparatively lower model complexity. The Stacking Regressor did not provide a significant improvement over the best individual ensemble models and was therefore not chosen for final deployment.

Performance of the Final Model on Test Data

- On the test dataset, the tuned Random Forest model demonstrates **high predictive reliability**, with shipment quantity predictions deviating by approximately **4% on average** from actual values.
- Such performance is considered **highly acceptable in FMCG supply chain planning**, where demand variability, logistics constraints, and operational uncertainties are inherent.

Important features of the Final Model

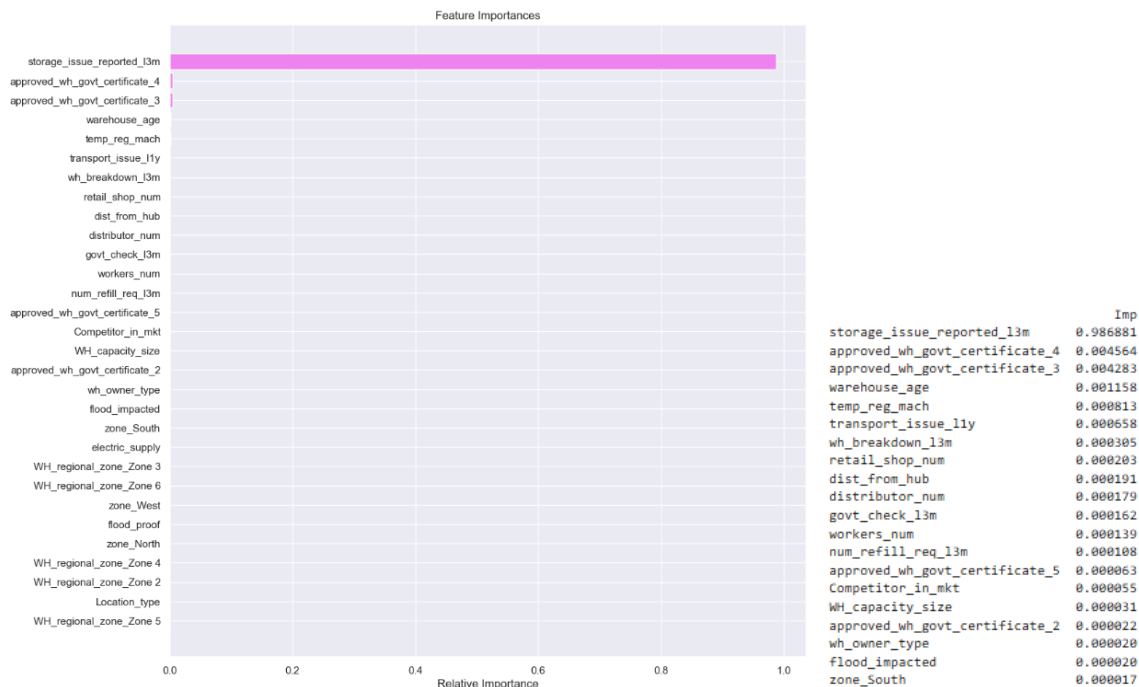


Figure 24: Important features of Final Model

Observation

The feature importance results from the tuned Random Forest model highlight the relative contribution of each predictor in estimating shipment quantity (product_wg_ton).

- storage_issue_reported_13m* overwhelmingly dominates the model, contributing ~98.7% of total importance. This indicates that the frequency of recent storage issues is the most critical driver of shipment volume. Warehouses reporting more storage issues are likely associated with higher throughput or corrective shipment adjustments, making this variable a strong proxy for operational intensity.
- Government certification levels, particularly *approved_wh_govt_certificate_4* and *approved_wh_govt_certificate_3*, emerge as the next most influential features. Although their individual importance values are much smaller than the top feature, they collectively indicate

that regulatory compliance and certification standards play a meaningful role in enabling higher shipment volumes.

- *warehouse_age* shows measurable importance, suggesting that older or more established warehouses tend to handle shipments differently compared to newer facilities, likely due to better infrastructure maturity and operational experience.
- Other operational, logistical, and regional variables contribute marginally, suggesting that shipment volumes are driven more by dynamic operational conditions than static warehouse attributes.

7. Actionable Insights & Recommendations

7.1 Insight

- **Operational intensity is the primary driver of shipment volume**, with recent storage activity emerging as the strongest indicator of demand and throughput requirements.
- **Regulatory readiness directly enables higher shipment capacity**, as warehouses with stronger government certification consistently support larger product movement.
- **Mature warehouses outperform newer facilities**, indicating that operational experience and infrastructure readiness play a critical role in meeting demand.
- **Advanced ensemble forecasting models deliver high reliability**, with the selected Random Forest model achieving approximately **4% prediction error**, making it suitable for enterprise-level supply planning.
- **Geographic and structural factors are secondary**, reinforcing the need to focus on dynamic operational signals rather than static warehouse attributes.

7.2 Business Recommendations

- **Adopt predictive shipment planning** using the Random Forest model to proactively align supply with warehouse-level demand and reduce inventory mismatches.
- **Strengthen operational governance in high-activity warehouses**, prioritizing preventive maintenance and storage optimization to support sustained throughput.
- **Accelerate compliance upgrades and certification improvements**, enabling warehouses to scale shipments efficiently and reduce regulatory constraints.
- **Invest selectively in newer or lower-maturity warehouses**, focusing on infrastructure, process standardization, and workforce capability development.
- **Shift from reactive to data-driven supply chain decisions**, using model-based demand insights to guide inventory allocation, production planning, and regional distribution strategies.

Final Recommendation

- Implement the tuned Random Forest model as a decision-support tool to guide warehouse-level shipment planning, reduce inventory mismatch, and enhance overall supply chain efficiency.

Thank You

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